1-Creating necessary users and groups and creating our directories

```
[root@server3 ~]# groupadd supervisors
[root@server3 ~]# groupadd engineers
[root@server3 ~]# gropuadd operators
bash: gropuadd: command not found...
Similar command is: 'groupadd'
[root@server3 ~]# groupadd operators
[root@server3 ~]# groupadd employees
[root@server3 ~]#
```

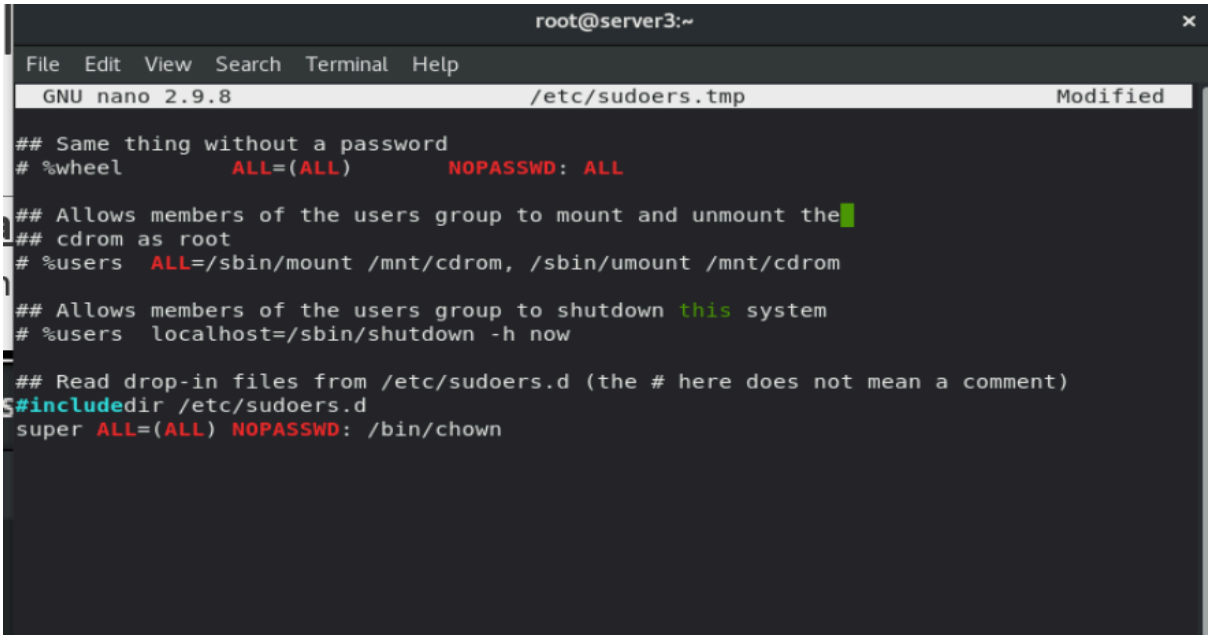
```
[root@server3 ~]# useradd super
[root@server3 ~]# passwd super
Changing password for user super.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server3 ~]# user eng1
bash: user: command not found...
[root@server3 ~]# useradd eng1
useradd: warning: the home directory already exists.
Not copying any file from skel directory into it.
Creating mailbox file: File exists
[root@server3 ~]# passwd eng1
Changing password for user eng1.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server3 ~]# useradd eng2
[root@server3 ~]# passwd eng2
Changing password for user eng2.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server3 ~]# useradd eng3
[root@server3 ~]# passwd eng3
Changing password for user eng3.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server3 ~]# useradd op1
[root@server3 ~]# passwd op1
Changing password for user op1.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server3 ~]# useradd op2
[root@server3 ~]# passwd op2
Changing password for user op2.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server3 ~]# useradd op3
[root@server3 ~]# passwd op3
Changing password for user op3.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
```

```
[root@server3 ~]# usermod -aG supervisors super
[root@server3 ~]# usermod -aG engineers,employees eng1
[root@server3 ~]# usermod -aG engineers,employees eng2
[root@server3 ~]# usermod -aG engineers,employees eng3
[root@server3 ~]# usermod -aG operators,employees op1
[root@server3 ~]# usermod -aG operators,employees op2
[root@server3 ~]# usermod -aG operators,employees op3
[root@server3 ~]# mkdir -p /factory
[root@server3 ~]# chown super:employees /factory
[root@server3 ~]# chmod 750 /factory
```

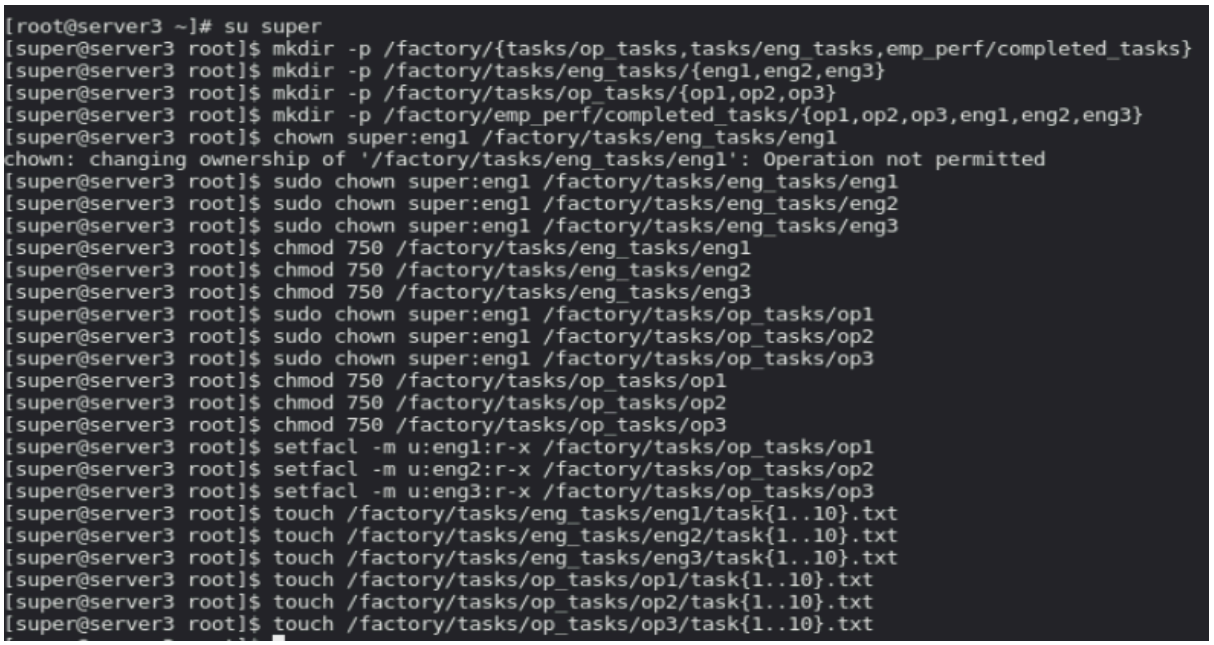
```
[root@server3 ~]# su super
[super@server3 root]$ mkdir -p /factory/{tasks/op_tasks,tasks/eng_tasks,emp_perf/completed_tasks}
[super@server3 root]$ mkdir -p /factory/tasks/eng_tasks/{eng1,eng2,eng3}
[super@server3 root]$ mkdir -p /factory/tasks/op_tasks/{op1,op2,op3}
[super@server3 root]$ mkdir -p /factory/emp_perf/completed_tasks/{op1,op2,op3,eng1,eng2,eng3}
[super@server3 root]$
```

2-Enabling super to use the chown command by adding him to the sudoers

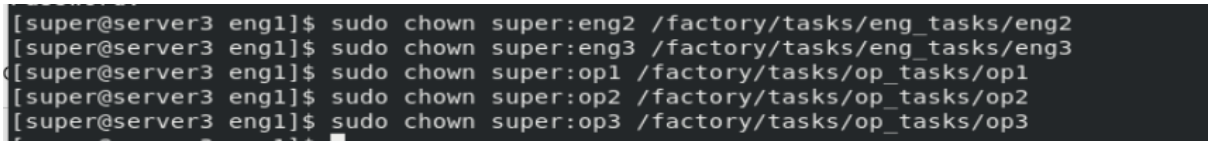
```
[student@server3 ~]$ su -
Password:
[root@server3 ~]# sude EDITOR=nano visudo
bash: sude: command not found...
[root@server3 ~]# sudo EDITOR=nano visudo
[root@server3 ~]#
```



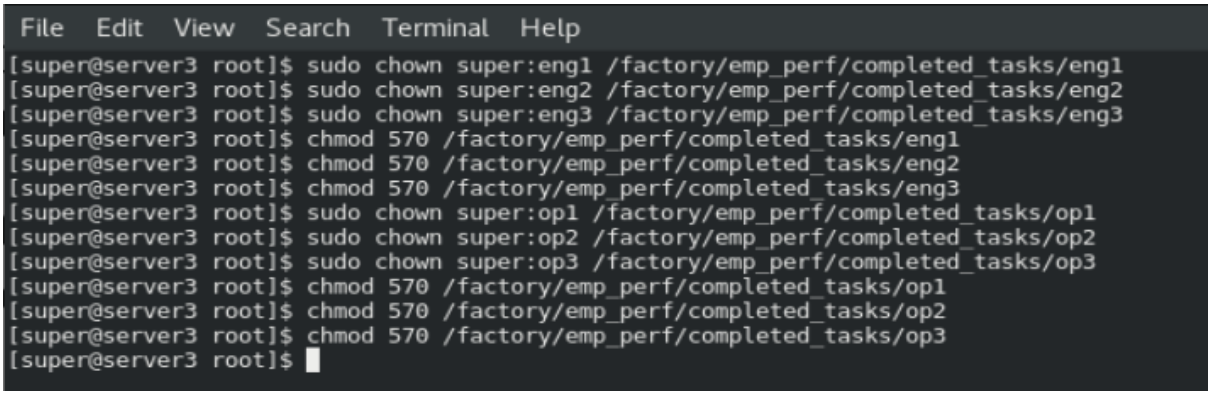
3-Creating the subdrirectories and setting permissions (there way a typo i handled it in the second photo) :



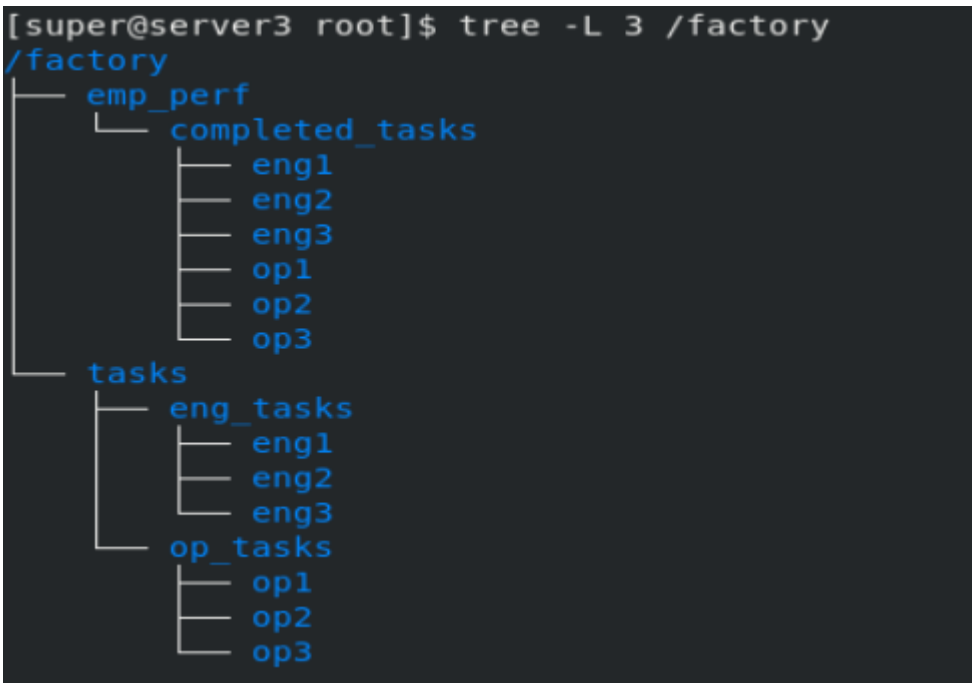
```
[root@server3 ~]# su super
[super@server3 root]$ mkdir -p /factory/{tasks/op_tasks,tasks/eng_tasks,emp_perf/completed_tasks}
[super@server3 root]$ mkdir -p /factory/tasks/eng_tasks/{eng1,eng2,eng3}
[super@server3 root]$ mkdir -p /factory/tasks/op_tasks/{op1,op2,op3}
[super@server3 root]$ mkdir -p /factory/emp_perf/completed_tasks/{op1,op2,op3,eng1,eng2,eng3}
[super@server3 root]$ chown super:eng1 /factory/tasks/eng_tasks/eng1
chown: changing ownership of '/factory/tasks/eng_tasks/eng1': Operation not permitted
[super@server3 root]$ sudo chown super:eng1 /factory/tasks/eng_tasks/eng1
[super@server3 root]$ sudo chown super:eng1 /factory/tasks/eng_tasks/eng2
[super@server3 root]$ sudo chown super:eng1 /factory/tasks/eng_tasks/eng3
[super@server3 root]$ chmod 750 /factory/tasks/eng_tasks/eng1
[super@server3 root]$ chmod 750 /factory/tasks/eng_tasks/eng2
[super@server3 root]$ chmod 750 /factory/tasks/eng_tasks/eng3
[super@server3 root]$ sudo chown super:eng1 /factory/tasks/op_tasks/op1
[super@server3 root]$ sudo chown super:eng1 /factory/tasks/op_tasks/op2
[super@server3 root]$ sudo chown super:eng1 /factory/tasks/op_tasks/op3
[super@server3 root]$ chmod 750 /factory/tasks/op_tasks/op1
[super@server3 root]$ chmod 750 /factory/tasks/op_tasks/op2
[super@server3 root]$ chmod 750 /factory/tasks/op_tasks/op3
[super@server3 root]$ setfacl -m u:eng1:r-x /factory/tasks/op_tasks/op1
[super@server3 root]$ setfacl -m u:eng2:r-x /factory/tasks/op_tasks/op2
[super@server3 root]$ setfacl -m u:eng3:r-x /factory/tasks/op_tasks/op3
[super@server3 root]$ touch /factory/tasks/eng_tasks/eng1/task{1..10}.txt
[super@server3 root]$ touch /factory/tasks/eng_tasks/eng2/task{1..10}.txt
[super@server3 root]$ touch /factory/tasks/eng_tasks/eng3/task{1..10}.txt
[super@server3 root]$ touch /factory/tasks/op_tasks/op1/task{1..10}.txt
[super@server3 root]$ touch /factory/tasks/op_tasks/op2/task{1..10}.txt
[super@server3 root]$ touch /factory/tasks/op_tasks/op3/task{1..10}.txt
```



```
[super@server3 eng1]$ sudo chown super:eng2 /factory/tasks/eng_tasks/eng2
[super@server3 eng1]$ sudo chown super:eng3 /factory/tasks/eng_tasks/eng3
[super@server3 eng1]$ sudo chown super:op1 /factory/tasks/op_tasks/op1
[super@server3 eng1]$ sudo chown super:op2 /factory/tasks/op_tasks/op2
[super@server3 eng1]$ sudo chown super:op3 /factory/tasks/op_tasks/op3
```



```
File  Edit  View  Search  Terminal  Help
[super@server3 root]$ sudo chown super:eng1 /factory/emp_perf/completed_tasks/eng1
[super@server3 root]$ sudo chown super:eng2 /factory/emp_perf/completed_tasks/eng2
[super@server3 root]$ sudo chown super:eng3 /factory/emp_perf/completed_tasks/eng3
[super@server3 root]$ chmod 570 /factory/emp_perf/completed_tasks/eng1
[super@server3 root]$ chmod 570 /factory/emp_perf/completed_tasks/eng2
[super@server3 root]$ chmod 570 /factory/emp_perf/completed_tasks/eng3
[super@server3 root]$ sudo chown super:op1 /factory/emp_perf/completed_tasks/op1
[super@server3 root]$ sudo chown super:op2 /factory/emp_perf/completed_tasks/op2
[super@server3 root]$ sudo chown super:op3 /factory/emp_perf/completed_tasks/op3
[super@server3 root]$ chmod 570 /factory/emp_perf/completed_tasks/op1
[super@server3 root]$ chmod 570 /factory/emp_perf/completed_tasks/op2
[super@server3 root]$ chmod 570 /factory/emp_perf/completed_tasks/op3
[super@server3 root]$
```


[super@server3 root]$ tree -L 3 /factory
/factory
emp_perf
completed_tasks
eng1
eng2
eng3
op1
op2
op3
tasks
eng_tasks
eng1
eng2
eng3
op_tasks
op1
op2
op3

4-the first script:

```
[super@server3 factory]$ mkdir /factory/emp_perf/scripts_for_employees
[super@server3 factory]$ cd /factory/emp_perf/scripts_for_employees
[super@server3 scripts_for_employees]$ nano submit_task.sh
[super@server3 scripts_for_employees]$ chown super:employees /factory/emp_perf/scripts_for_employees
chown: changing ownership of '/factory/emp_perf/scripts_for_employees': Operation not permitted
[super@server3 scripts_for_employees]$ sudo chown super:employees /factory/emp_perf/scripts_for_employees
[super@server3 scripts_for_employees]$ chmod 750 /factory/emp_perf/scripts_for_employees
```

```
[super@server3 scripts_for_employees]$ sudo chown super:employees /factory/emp_perf/scripts_for_employees/submit_task.sh
[super@server3 scripts_for_employees]$ chmod 750 /factory/emp_perf/scripts_for_employees/submit_task.sh
[super@server3 scripts_for_employees]$
```

```
  GNU nano 2.9.8                                                    /fac

#!/bin/bash
file="$1"

if groups "$USER" | grep -q '\bengineers\b'; then
task="/factory/tasks/eng_tasks"
else
task="/factory/tasks/op_tasks"
fi

completed="/factory/emp_perf/completed_tasks"
user="$USER"
if [ -z "$file" ]; then
        echo "please provide the task completed"
exit 1
fi
if [ "$#" -ne 1 ]; then
echo "only submit one task at a time "
exit 1
fi
if [ ! -f "$task/${user}/${file}" ]; then
echo "file '$file' does not exist in your tasks directory"
exit 1
fi

if [ -f "$completed/${user}/${file}" ]; then
echo "you already submitted this task"
exit 1
fi

cp "$task/${user}/${file}" "$completed/${user}/"

if [ $? -eq 0 ]; then
echo "task submitted"

else "error submitting the task pleae try again"
exit 1
fi
```

```
[super@server3 eng3]$ su op1
Password:
[op1@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task4
file 'task4' does not exist in your tasks directory
[op1@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task4.txt
task submitted
[op1@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task4.txt
you already submitted this task
[op1@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task4.txt dd
d
only submit one task at a time
[op1@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh talll
file 'talll' does not exist in your tasks directory
[op1@server3 eng3]$
```

```
[eng3@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task7.txt
task submitted
[eng3@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task7.txt
you already submitted this task
[eng3@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task16.txt
file 'task16.txt' does not exist in your tasks directory
[eng3@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh task16.txt dddd
only submit one task at a time
[eng3@server3 eng3]$ /factory/emp_perf/scripts_for_employees/submit_task.sh
please provide the task completed
```

```
[super@server3 eng3]$ tree /factory/emp_perf
/factory/emp_perf
├── completed_tasks
│   ├── eng1
│   ├── eng2
│   ├── eng3
│   │   └── task7.txt
│   ├── op1
│   │   └── task4.txt
│   ├── op2
│   └── op3
└── scripts_for_employees
    └── submit_task.sh

8 directories, 3 files
```

```
[root@server3 ~]# su super
[super@server3 root]$ tree /factory/emp_perf/completed_tasks
/factory/emp_perf/completed_tasks
├── eng1
│   └── task9.txt
├── eng2
│   └── task1.txt
├── eng3
│   └── task7.txt
├── op1
│   └── task4.txt
├── op2
│   └── task6.txt
└── op3
    └── task1.txt

6 directories, 6 files
[super@server3 root]$
```

9-the second script:

```
[super@server3 ~]$ cd /factory/emp_perf
[super@server3 emp_perf]$ ls
completed_tasks  scripts_for_employees  super_scripts
[super@server3 emp_perf]$ cd super_scripts
[super@server3 super_scripts]$ ls
[super@server3 super_scripts]$ nano calc_kpi.sh
```

```
[super@server3 super_scripts]$ chown super:super /factory/emp_perf/super_scripts/calc_k
pi.sh
[super@server3 super_scripts]$ chmod 770 /factory/emp_perf/super_scripts/calc_kpi.sh
```

super@server3:~

GNU nano 2.9.8 /factory/emp_perf/

```
tasks="/factory/tasks"
completed="/factory/emp_perf/completed_tasks"
date=$(date "+%Y-%m-%d %H:%M:%S")

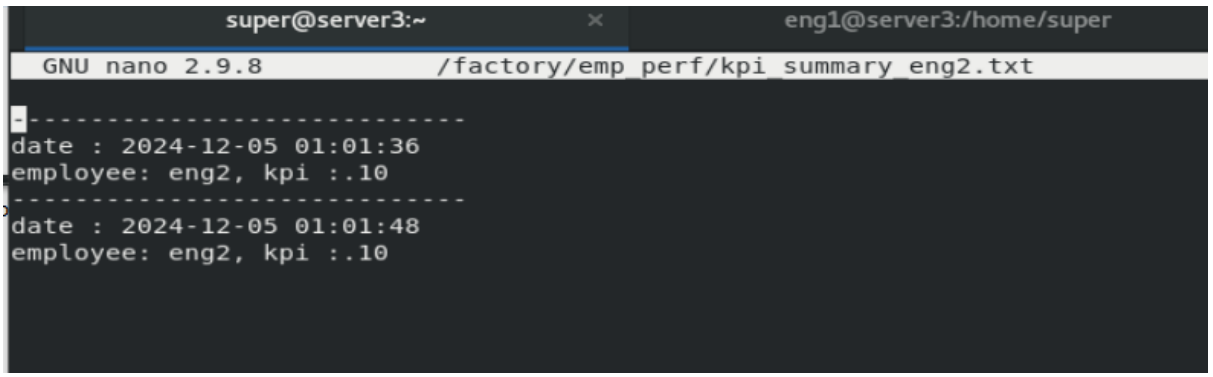
for user in "$completed"/*; do
        user_name=$(basename "$user")
        file="/factory/emp_perf/kpi_summary_${user_name}.txt"

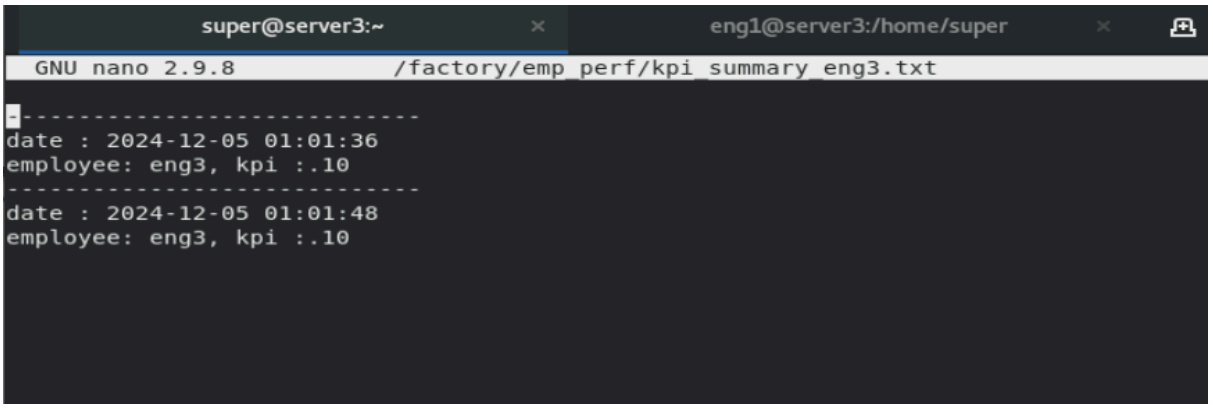
        if [ ! -f "$file" ]; then
                touch "$file"
                echo "created kpi file for the first time for ${user_name} "
                chown super:supervisors "$file"
                chmod 770 "$file"
        fi

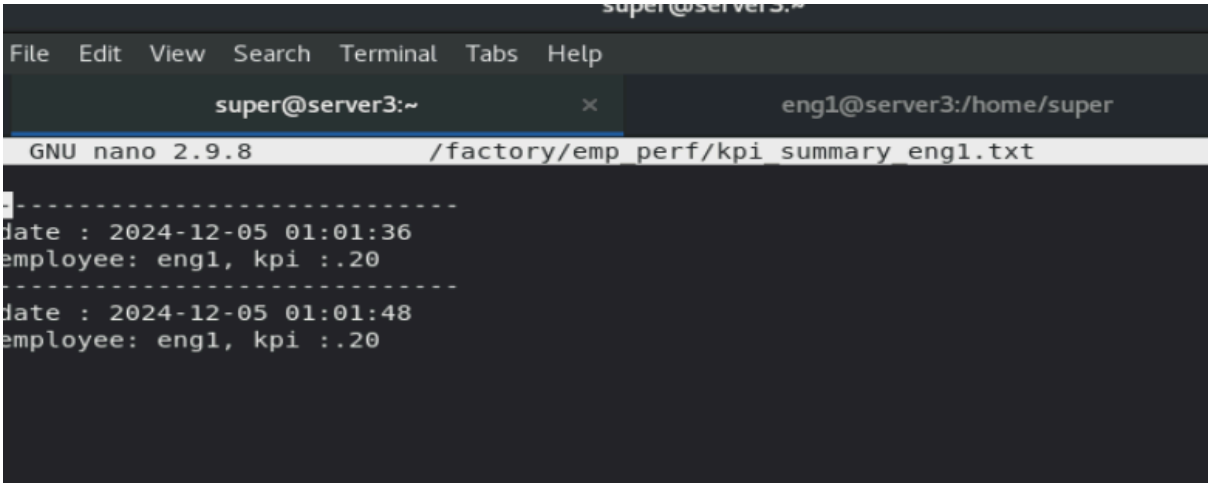
        echo "----------------------------" >> "$file"
        echo "date : $date" >> "$file"

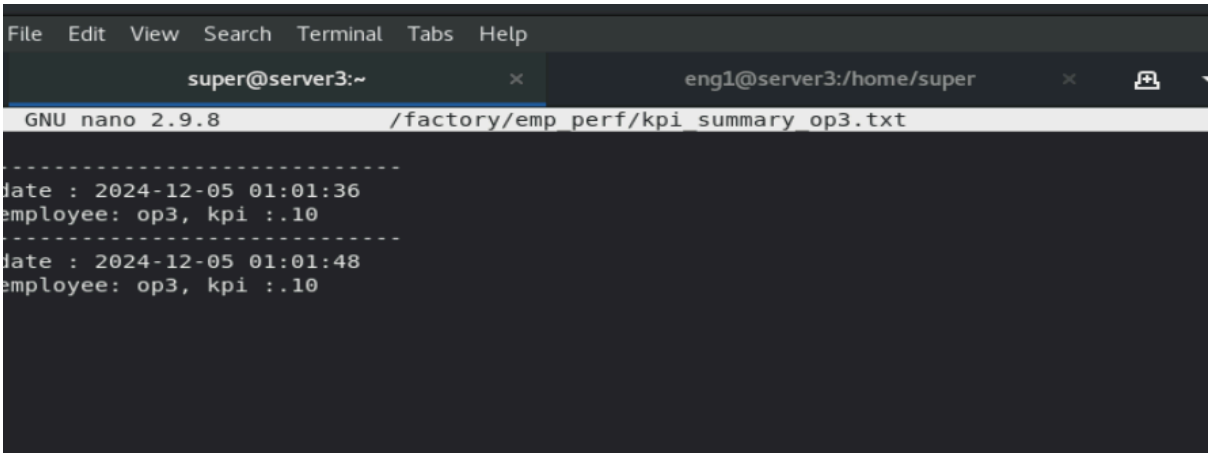

        if [ -d "$tasks/eng_tasks/$user_name" ]; then
                tasks_dir="$tasks/eng_tasks/$user_name"
        else
                tasks_dir="$tasks/op_tasks/$user_name"
        fi
        assigned=$(find "$tasks_dir" -type f | wc -l)
        submitted=0
        for task in "$user"/*; do
                if [ -f "$task" ]; then
                        task_name=$(basename "$task")
                        if [ -f "$tasks_dir/$task_name" ]; then
                                submitted=$((submitted+1))
                        fi
                fi
        done
        if [ "$submitted" -eq 0 ]; then
                echo "no assigned tasks for $user_name"
                continue
        fi
        kpi=$(echo "scale=2; $submitted / $assigned" | bc)
        echo "employee: $user_name, kpi :$kpi" >> "$file"
done
```

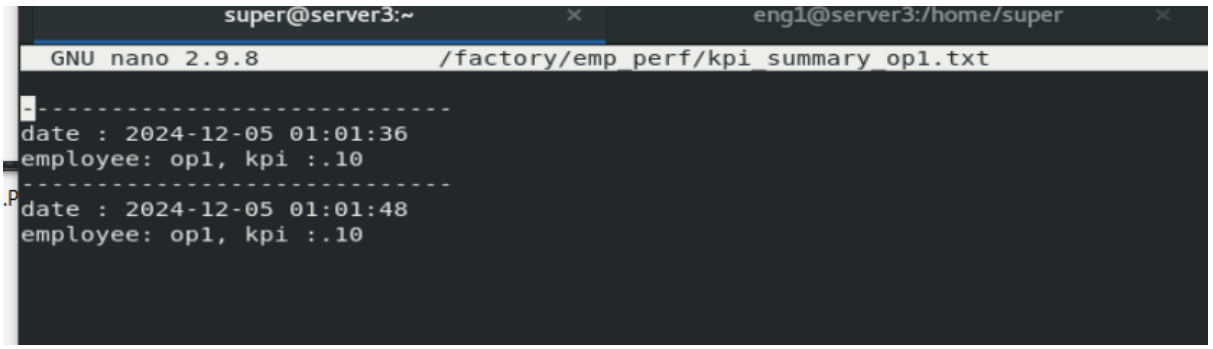
```
[super@server3 ~]$ nano /factory/emp_perf/super_scripts/calc_kpi.sh
[super@server3 ~]$ /factory/emp_perf/super_scripts/calc_kpi.sh
created kpi file for the first time for eng1
created kpi file for the first time for eng2
created kpi file for the first time for eng3
created kpi file for the first time for op1
created kpi file for the first time for op2
created kpi file for the first time for op3
all kpis have been calculated
[super@server3 ~]$ /factory/emp_perf/super_scripts/calc_kpi.sh
all kpis have been calculated
[super@server3 ~]$
```

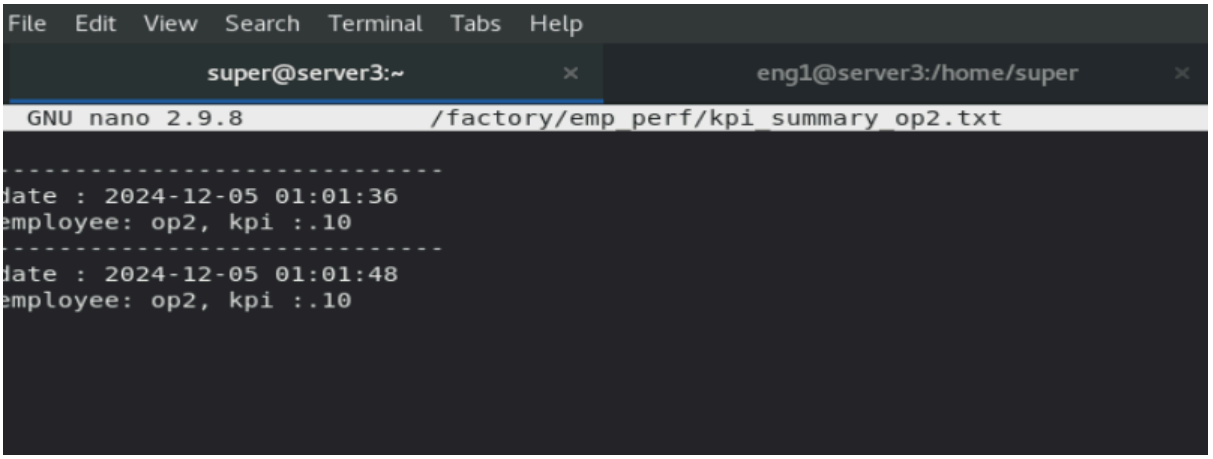

super@server3:~
eng1@server3:/home/super
GNU nano 2.9.8 /factory/emp_perf/kpi_summary_eng2.txt
date : 2024-12-05 01:01:36
employee: eng2, kpi :.10
date : 2024-12-05 01:01:48
employee: eng2, kpi :.10


super@server3:~
eng1@server3:/home/super
GNU nano 2.9.8 /factory/emp_perf/kpi_summary_eng3.txt
date : 2024-12-05 01:01:36
employee: eng3, kpi :.10
date : 2024-12-05 01:01:48
employee: eng3, kpi :.10


File Edit View Search Terminal Tabs Help
super@server3:~
eng1@server3:/home/super
GNU nano 2.9.8 /factory/emp_perf/kpi_summary_eng1.txt
date : 2024-12-05 01:01:36
employee: eng1, kpi :.20
date : 2024-12-05 01:01:48
employee: eng1, kpi :.20


File Edit View Search Terminal Tabs Help
super@server3:~
eng1@server3:/home/super
GNU nano 2.9.8 /factory/emp_perf/kpi_summary_op3.txt
date : 2024-12-05 01:01:36
employee: op3, kpi :.10
date : 2024-12-05 01:01:48
employee: op3, kpi :.10


super@server3:~
eng1@server3:/home/super
GNU nano 2.9.8 /factory/emp_perf/kpi_summary_op1.txt
----
date : 2024-12-05 01:01:36
employee: op1, kpi :.10
----
date : 2024-12-05 01:01:48
employee: op1, kpi :.10


File Edit View Search Terminal Tabs Help
super@server3:~
eng1@server3:/home/super
GNU nano 2.9.8 /factory/emp_perf/kpi_summary_op2.txt
----
date : 2024-12-05 01:01:36
employee: op2, kpi :.10
----
date : 2024-12-05 01:01:48
employee: op2, kpi :.10